AE 153 – Agricultural Meteorology & Climatology (2,1,2)

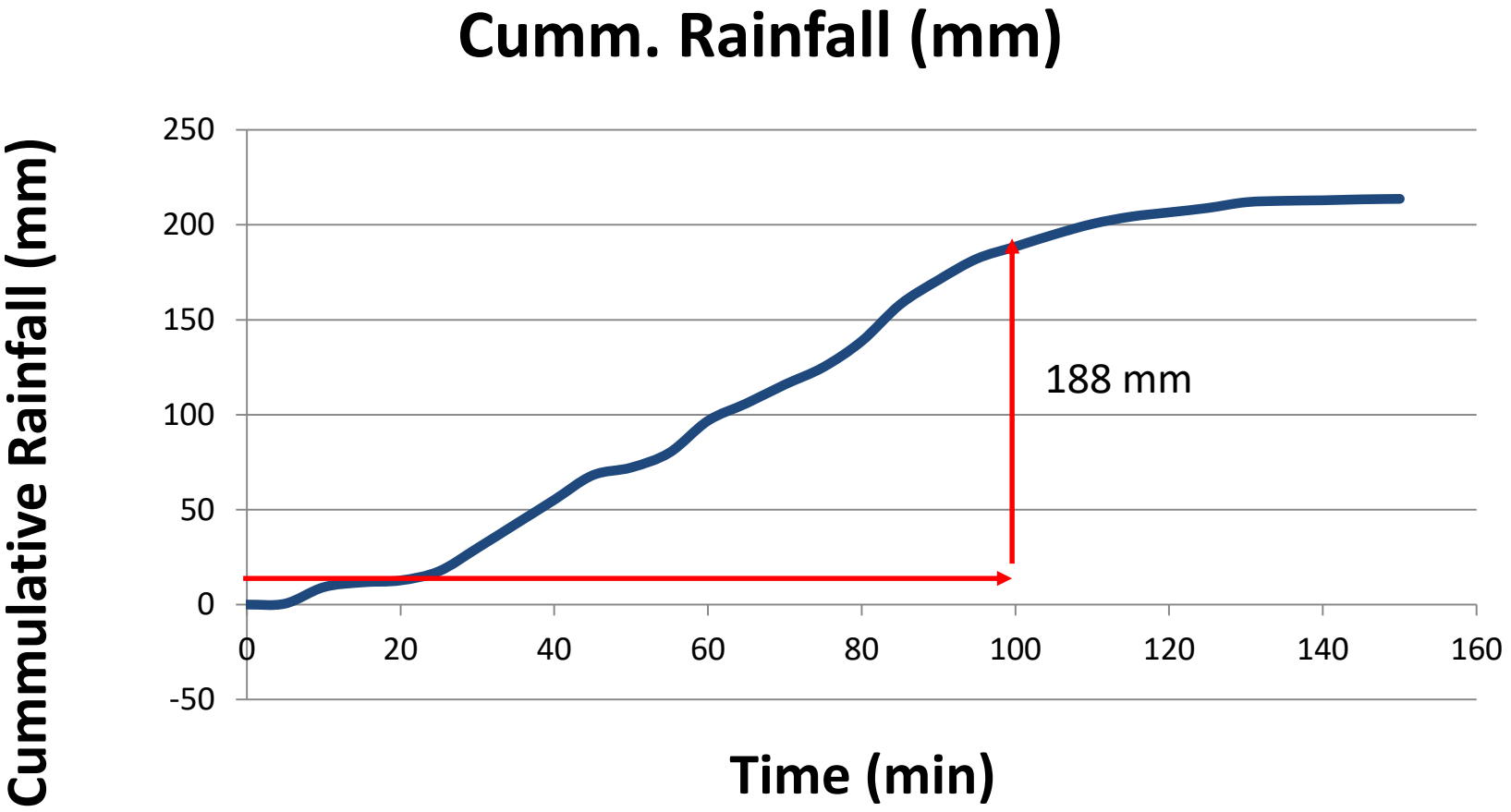
E. Bessah

Data Presentation and Processing

Rainfall data presentation

1. Cumulative mass curve

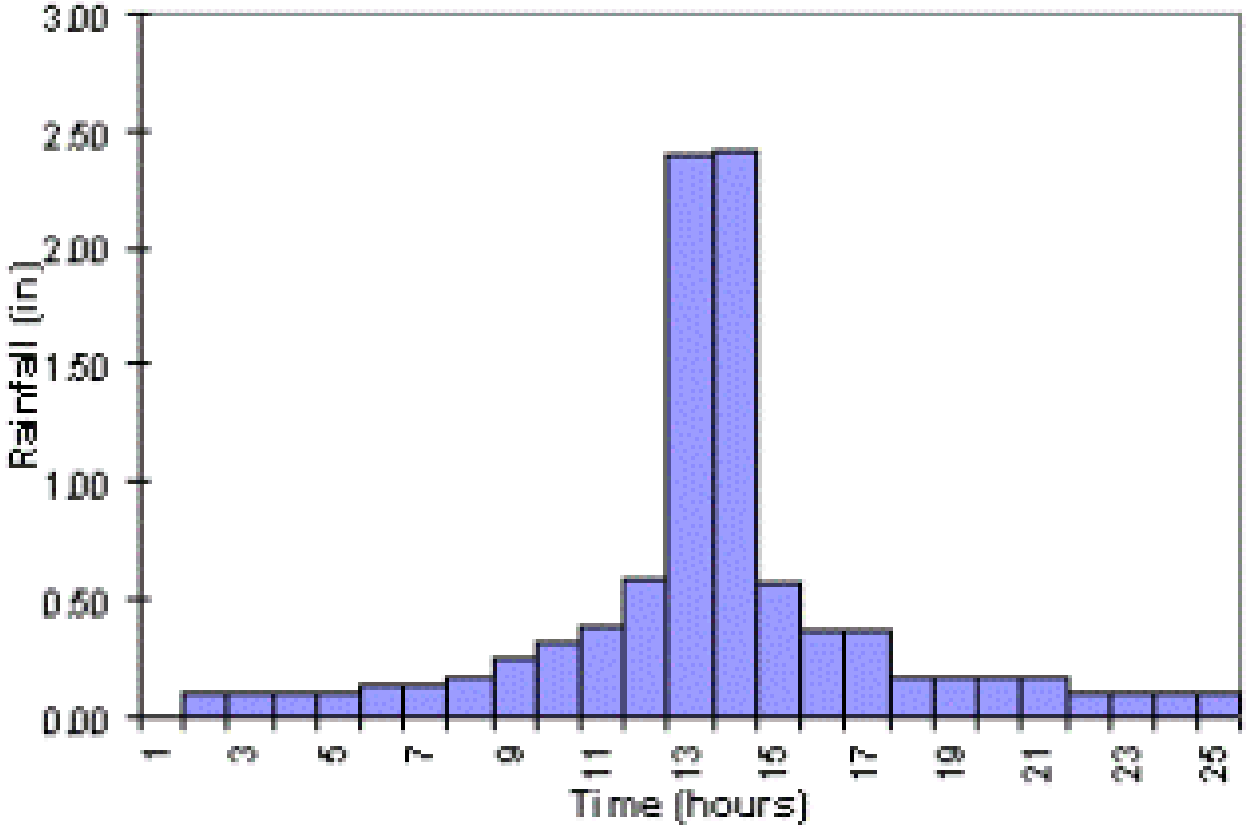
Plot of summation of rainfall increments as a function of time
Typically the output from a recording gauge is in the form of cumulative rainfall over time.



Rainfall data presentation

2. Rainfall hyetograph

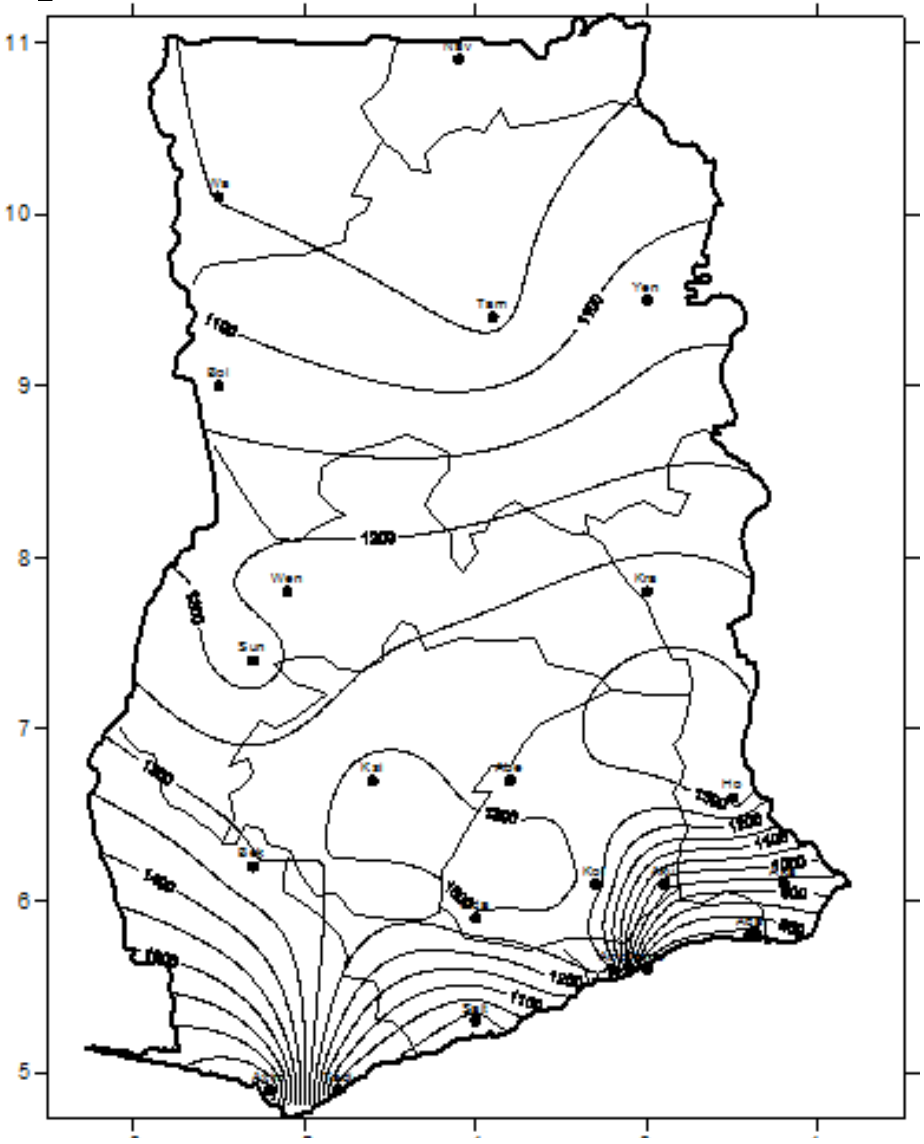
The depth or intensity of rainfall plotted against time interval to generate a hyetograph characterizing the time distribution of a storm or selected time period. Important input to hydrologic models (surface hydrology models).



Rainfall data presentation

3. **Isohyetal maps**

   Isohyetal map is a plot of constant rainfall depth/amount (sometimes intensity) contours based on rainfall data recorded at gauged locations (points). These are generated similarly to topographic contour maps. Isohyetal maps are used to represent long-term averages or individual storm events.



Isoyetal map of Ghana

Point rainfall data assessment

We measure rainfall at a point which is used for the following:

- Estimate average rainfall over an area (watershed)
- Transform point rainfall to aerial rainfall for model applications to large areas
- Estimate variation in rainfall data in terms of space or time
- Estimate missing record of gauge data or estimate data at an un-gauged location

Estimation of Precipitation over an Area

Methods: Arithmetic average/mean; Thiessen polygon; and Isoyetal

1. **Arithmetic Averages***: Used when*
   - *gauges are uniformly distributed*
   - *individual gauge measurements do not vary greatly (typically for small areas)*

$$Pavg = \frac{\sum_{i=1}^{n} P_i}{n}$$

P_i = rainfall at gauge or location **i**

n = number of gauges or locations

2. Thiessen Polygon

- Underlying concept is that rainfall for an area is represented by the nearest gauge
- This is done by constructing polygons around each gauge, as follows:
  a. First, by drawing lines to connect each gauge.
  b. Then draw perpendicular bisector on each connecting line.
  c. Connect bisectors to form polygons
  d. Determine area for each polygon

Thiessen Polygon

$$Pavg = \frac{\sum_{i=1}^{n} A_i P_i}{A_T}$$

n = number of polygons (gauges or locations)

A_T = total area

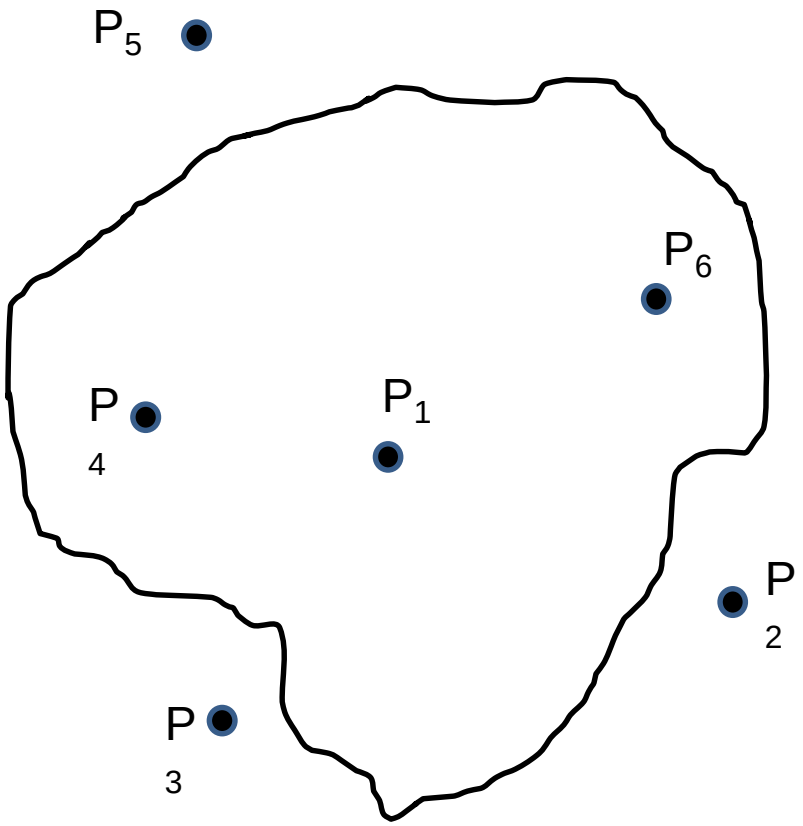
A_i = area of polygon around gauge or location i

P_i = rainfall at gauge or location i

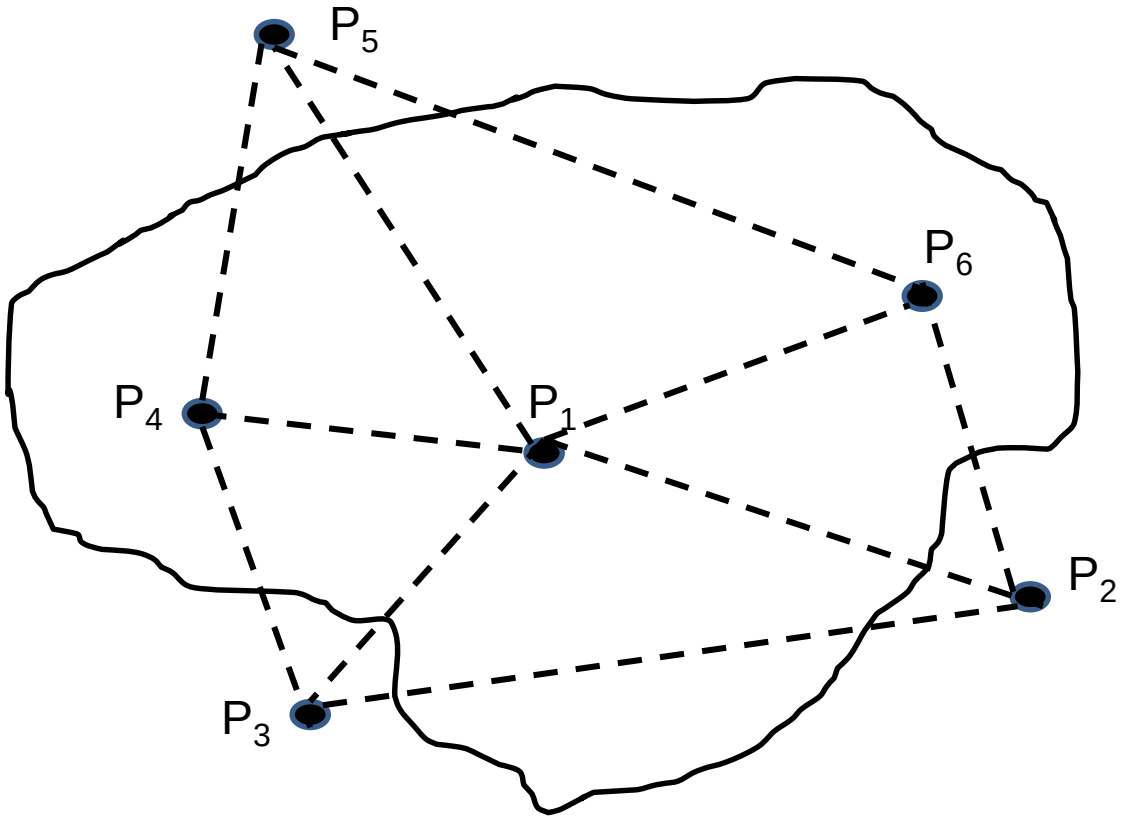
Sample calculation: Arithmetic Average and Thiessen polygon

Determine the mean rainfall using the arithmetic average and the Thiessen polygon methods

Rain gauge	**Precipitation (P_i)(mm)**
P_1	56
P_2	40
P_3	44
P_4	52
P_5	48
P_6	52
Total	

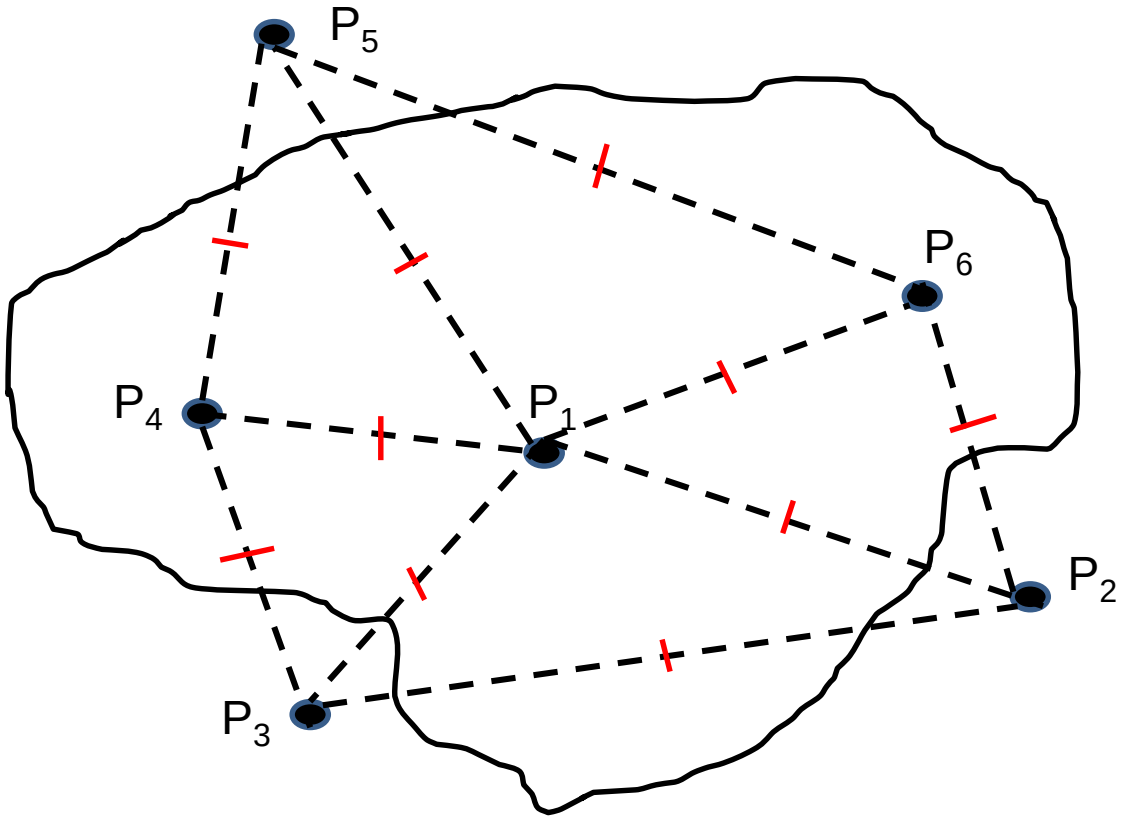


Thiessen polygon



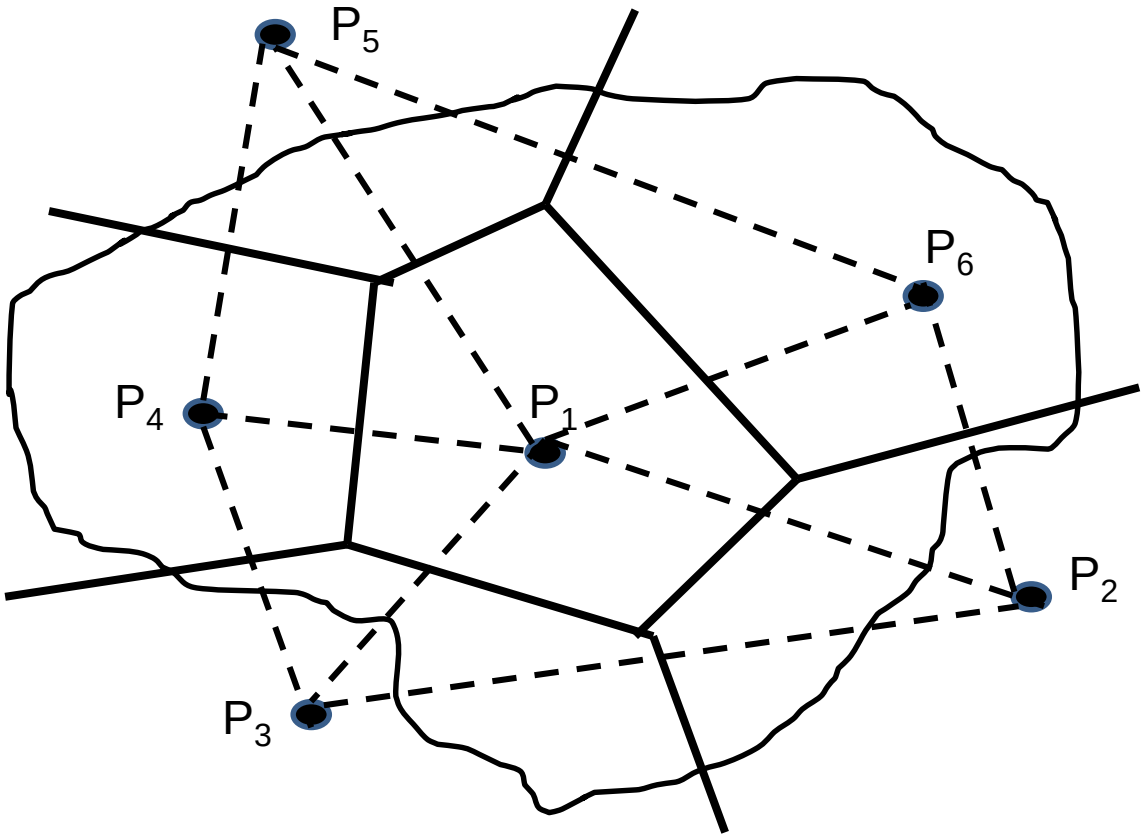
a. Drawing lines to connect each gauge

Thiessen polygon



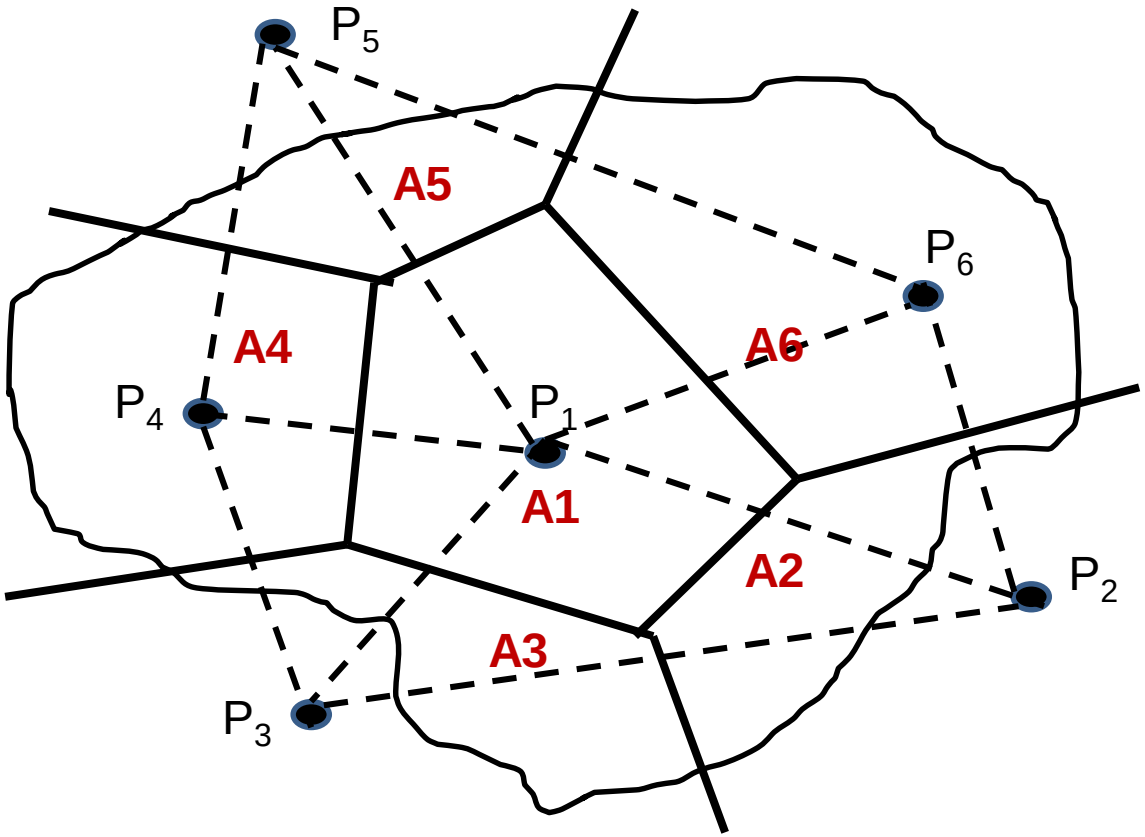
b. Then draw perpendicular bisector on each connecting line

Thiessen polygon



c. Connect bisectors to form polygons

Thiessen polygon



d. Determine area for each polygon

Arithmetic Average and Thiessen polygon

Rain gauge	**Precipitation (P_i)(mm)**	**Area of influence (A_i)(ha)**	**P_iA_i**
P_1	56	260	14560
P_2	40	58	2320
P_3	44	50	2200
P_4	52	250	13000
P_5	48	62	2976
P_6	52	290	15080
Total		970	50136

$$Pavg = \frac{\sum_{i=1}^{n} P_i}{n}$$

$$Av = \frac{292}{6} = 48.67$$

$$Pavg = \frac{\sum_{i=1}^{6} A_i P_i}{A_T}$$

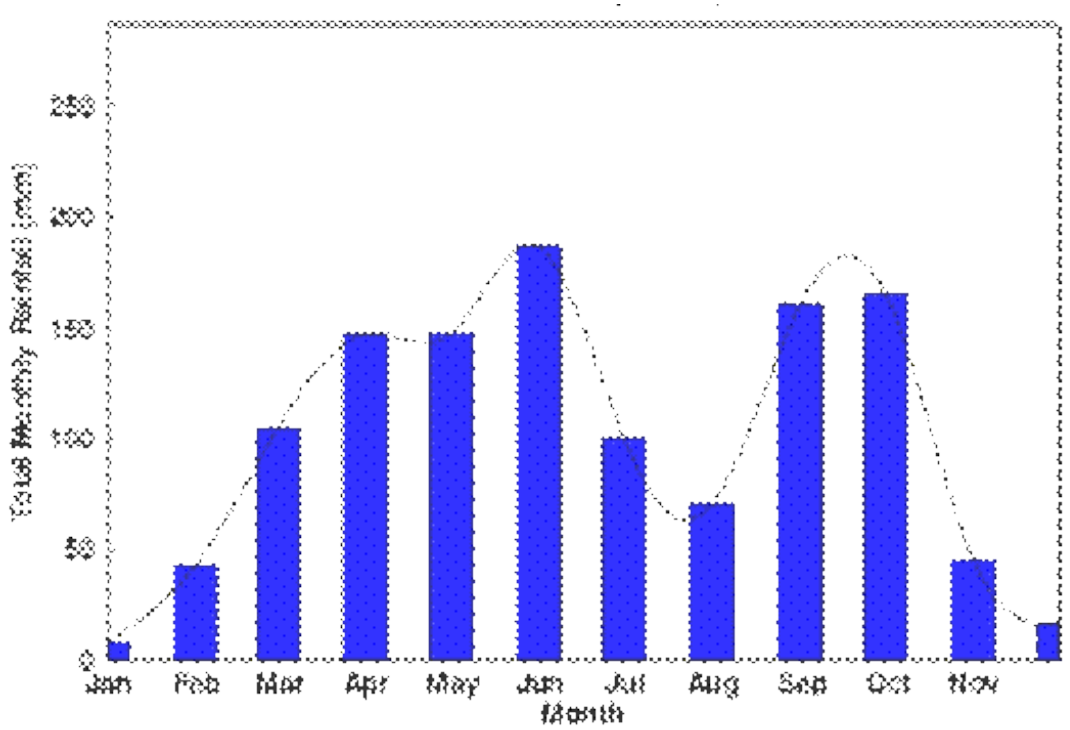
$$Av = \frac{50136}{970} = 51.69\ mm$$

3. *Isohyetal Method*

- Construct isohyets between gauges
- Calculate area between isohyets
- Estimate average rainfall between isohyetals and apply the same summation formula as in Thiessen Polygon where A_i and P_i represent the area and average rain between two isohyets.

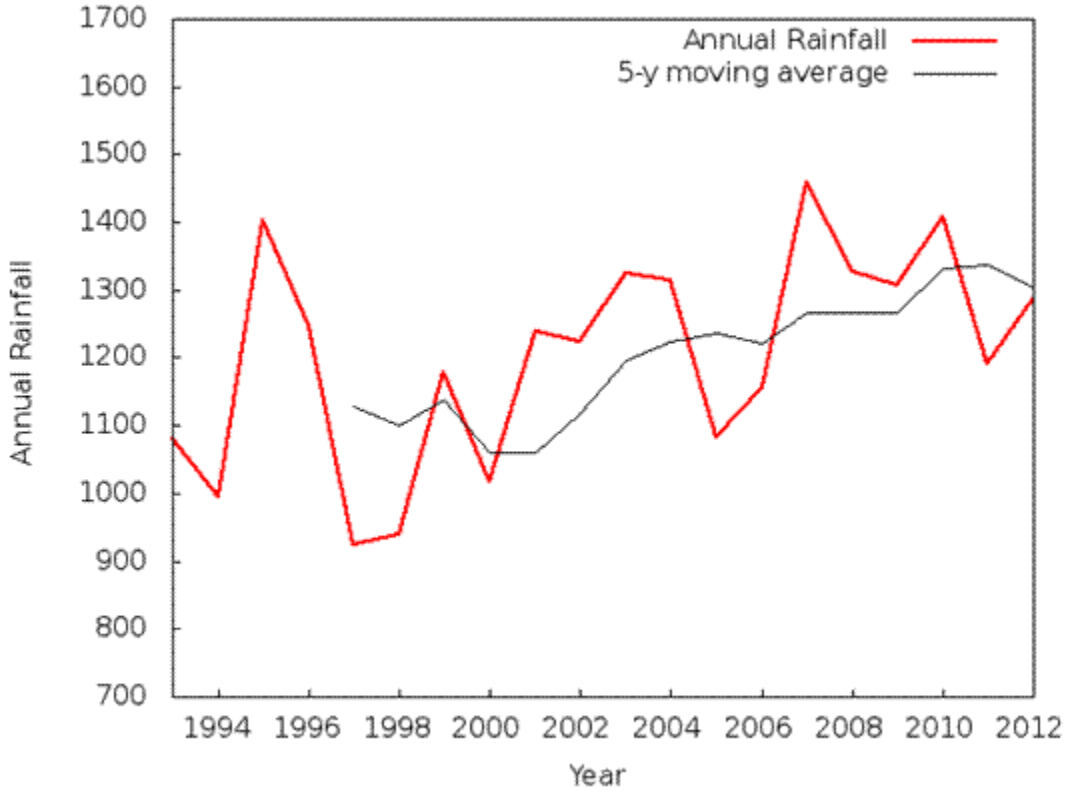
Rainfall Variability

Monthly rainfall distribution



Rainfall Variability

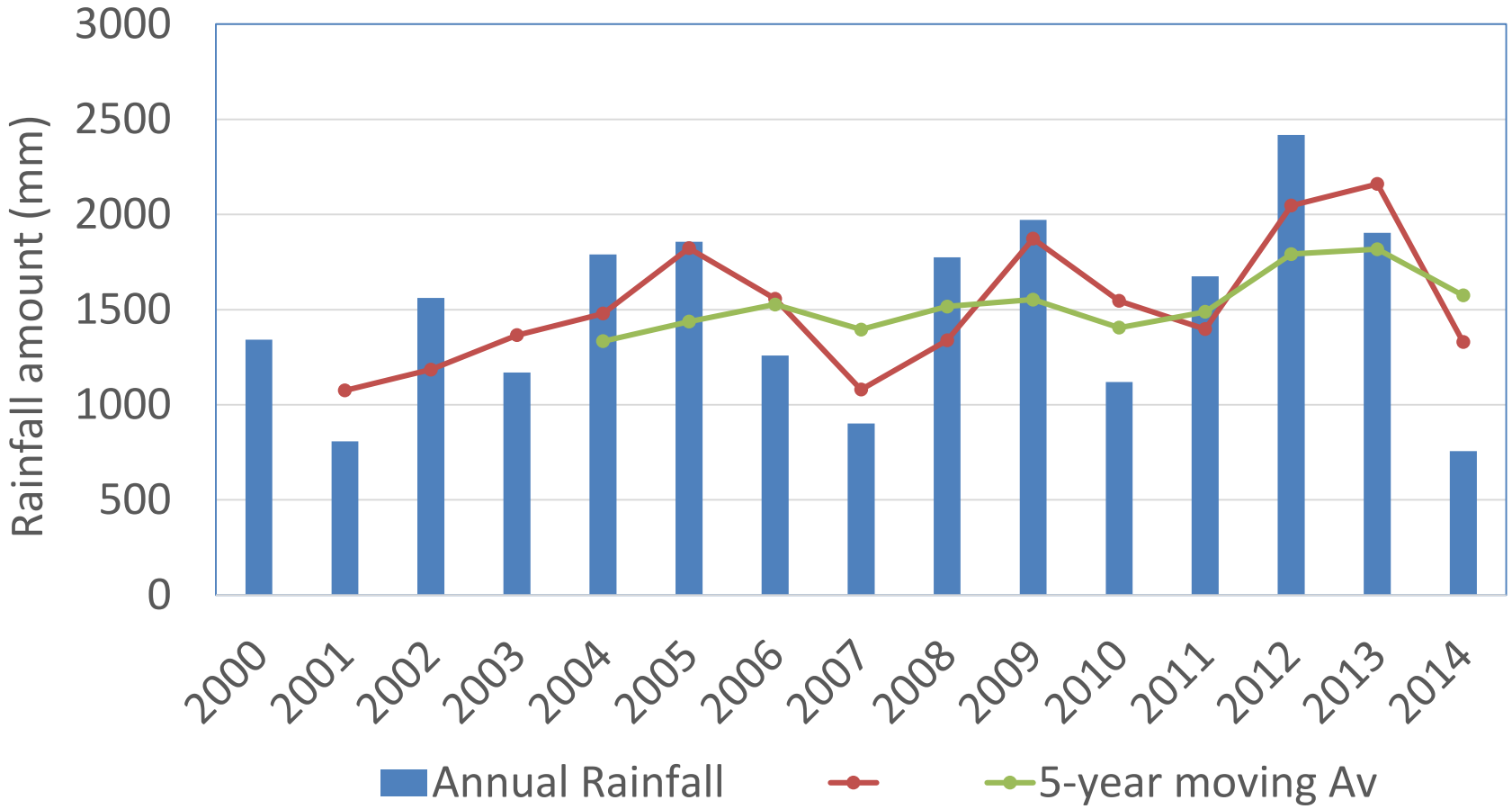
Rainfall trend analysis - Moving average i.e. average over number of months/years/events



Moving Average

SN	Year	Annual Rainfall	2-year moving Av	5-year moving Av	
1	2000	1342			
2	2001	808	1075		
3	2002	1562	1185		
4	2003	1169	1365.5		
5	2004	1790	1479.5	1334.2	
6	2005	1856	1823	1437	
7	2006	1258	1557	1527	
8	2007	902	1080	1395	
9	2008	1775	1338.5	1516.2	
10	2009	1971	1873	1552.4	
11	2010	1120	1545.5	1405.2	
12	2011	1675	1397.5	1488.6	
13	2012	2418	2046.5	1791.8	
14	2013	1903	2160.5	1817.4	
15	2014	756	1329.5	1574.4	
		1487			

Moving Average



Rainfall Variability

- **Mean** ($\bar{x}$) of data – average of the data

$$\bar{x} = \frac{\sum X_i}{n}$$

- **Standard deviation** (s) - measures the spread or variation of the individual rainfall data.

$$S = \sqrt{\frac{\sum (X_i - \dot{x})^2}{n-1}}$$

Rainfall Variability

Standard deviation

- The value of S is used to compare rainfall amounts variability for a given period for different stations.
- The smaller the value of S, the smaller the variation of rainfall.

Standard deviation estimation

A rough estimate of S is obtained by dividing the range (Highest minus lowest rainfall value) by a value depending on the number of data points

$$S = \frac{Xmax - Xmin}{n^*}$$

Number of data (n) closest to this given number	Divide range by (n*)
5	2
10	3
25	4
100	5

Rainfall Variability

- Coefficient of variation: measure of variation in data expressed as a percentage of the mean value. Useful in comparing rainfall variability of different stations

$$CV = \frac{s}{\bar{x}} \times 100$$

- CV < 0.20 or 20% → moderate variability and for most purposes an acceptable length of data series
- CV > 0.25 or 25% → either the series of data are too short for reliable estimates or the rainfall is highly variable

Sample calculation

Provided is May rainfall data for a 20 year period.
Estimate the mean, standard deviation and CV, and comment on your answer

SN	Year	x
1	1981	85
2	1982	90
3	1983	65
4	1984	50
5	1985	40
6	1986	28
7	1987	65
8	1988	40
9	1989	82
10	1990	35
11	1991	30
12	1992	70
13	1993	75
14	1994	45
15	1995	55
16	1996	80
17	1997	68
18	1998	44
19	1999	50
20	2000	72

Sample calculation

SN	Year	X	$(X - \dot{X})$	$(X - \dot{X})^2$
1	1981	85	26.55	704.90
2	1982	90	31.55	995.40
3	1983	65	6.55	42.90
4	1984	50	-8.45	71.40
5	1985	40	-18.45	340.40
6	1986	28	-30.45	927.20
7	1987	65	6.55	42.90
8	1988	40	-18.45	340.40
9	1989	82	23.55	554.60
10	1990	35	-23.45	549.90
11	1991	30	-28.45	809.40
12	1992	70	11.55	133.40
13	1993	75	16.55	273.90
14	1994	45	-13.45	180.90
15	1995	55	-3.45	11.90
16	1996	80	21.55	464.40
17	1997	68	9.55	91.20
18	1998	44	-14.45	208.80
19	1999	50	-8.45	71.40
20	2000	72	13.55	183.60

Sample calculation

- $\dot{X} = \frac{\sum x}{n} = \frac{85+90+65+\cdots+72}{20}$
- $\dot{X} = \frac{1169}{20} = 58.45$
- $S = \sqrt{\frac{\sum(X_i - \dot{x})^2}{n-1}} = \sqrt{\frac{6998.95}{19}} = 19.19$
- rough estimate of s = 15.15

- $CV = \frac{s}{\bar{x}} \times 100 = \frac{19.19}{58.45} \times 100 = 32.83\%$

Sample calculation

- Rainfall at the station is highly variable therefore the data may be unreliable and /or the period of observation is too short.
- Data should, if possible be correlated with data of nearby stations to check for validity.
- Also data should be checked for possible observation and instrument errors.

Assignment 4

i. Table 1 presents weekly rainfall, temperature, and cup anemometer reading collected at 0900 hours over a period of six (6) weeks in 2013 at DABE Weather Station.
ii. What is rainfall for the period
iii. What is mean weekly temperature
iv. What is mean temperature for the period
v. Calculate the standard deviation and CV for weekly rainfall and temperature
vi. Based on your answers in v, compare and comment on the variability in the weekly rainfall and temperature
vii. What is the highest weekly wind speed (km/h)
viii. Which week is likely to have the highest and lowest Relative Humidity and why

Assignment 4

Table 1 presents weekly rainfall, temperature, and cup anemometer

Week	Rainfall (mm)	Max Temp. (° C)	Min Temp. (° C)	Anemometer Reading (km)	Dry Bulb Temp. (° C)	Wet Bulb Temp. (° C)
1	25.4	29.3	21.9	5917.1	24.3	22.7
2	48.2	30.0	21.6	6446.5	23.8	22.4
3	3.2	29.1	21.5	7033.9	23.3	22.0
4	64.9	30.2	21.5	7462.5	23.9	22.6
5	39.4	30.1	21.3	7894.3	22.7	22.2
6	15.6	29.5	20.8	8412.4	23.5	22.3
7				8848.7		

Missing data point or rainfall at ungauged locations (1/6)

1. Simple Average of surrounding gauges

   Used for:

   i. Missing data estimation

   ii. Ungauged location data estimation

   Requirement/Conditions:

   a. Gauges must be of about equal distance from missing data point

   b. Gauges should have fairly equal data (e.g. rainfall) at known gauges

Missing data point or rainfall at ungauged locations (2/6)

2. Normal Ratio Method for missing data estimation
   - Stations are weighted based on normal annual data (precipitation) ratio's

   Used for:
   - Missing data estimation

   Requirement/Conditions :
   - When surrounding gauges differ by more than 10% in measured rainfall

Missing Data Point or Rainfall at Ungauged Locations (3/6)

Normal Ratio Method for missing data

$$P_x = \frac{N_x}{k}\left(\frac{P_1}{N_1} + \frac{P_2}{N_2} + ... + \frac{P_k}{N_k}\right)$$

Where:

N_x = normal long-term annual data (precipitation) at location x (i.e. location of interest)

P_k = event data (precipitation) at gauge k

N_k = normal long-term annual precipitation at gauge k

P_x = estimated event precipitation at location x (i.e. location of interest)

K = n - 1

The event can be a single storm, daily total, monthly total or annual value.

Sample Question

- Table 1 presents the monthly rainfall for June 2012 and long term mean annual rainfall for each of the rain gauge locations.
- Using the simple arithmetic average calculate mean monthly rainfall for June 2012 for the watershed
- Write an expression for the estimation of missing data using Normal Ratio Method
- Estimate the missing rainfall data for June 2012 at location P_6 using the expression and the data in Table …

Sample Question

Table 1 presents the monthly rainfall for June 2012

Rain gauge	Monthly rainfall for June 2012 (mm)	Long term annual rainfall (mm)
P_1	115	1300
P_2	156	1550
P_3	140	1350
P_4	150	1448
P_5	130	1288
P_6	X	1290
P_7	152	1524

Solution

Rain gauge	Monthly rainfall for June 2012 (mm) (Pi)	Long term annual rainfall (mm) (Ni)	P_k/N_k
P_1	115	1300	0.08846
P_2	156	1550	0.10065
P_3	140	1350	0.10370
P_4	150	1448	0.10359
P_5	130	1288	0.10093
P_6	X	1290	
P_7	152	1524	0.09974
Total			0.59707
Mean	**140.5**	Px	**128.37**

Sample Question

Estimate the missing rainfall data at location P_3

Rain gauge	Monthly rainfall for May 2014 (mm)	Long term annual rainfall (mm)
P_1	110	1200
P_2	89	900
P_3	X	1350
P_4	120	1458
P_5	95	1188
P_6	116	1145
P_7	122	1500

Missing data point or rainfall at ungauged locations (4/6)

3. Reciprocal (Inverse weighted) Distance Method

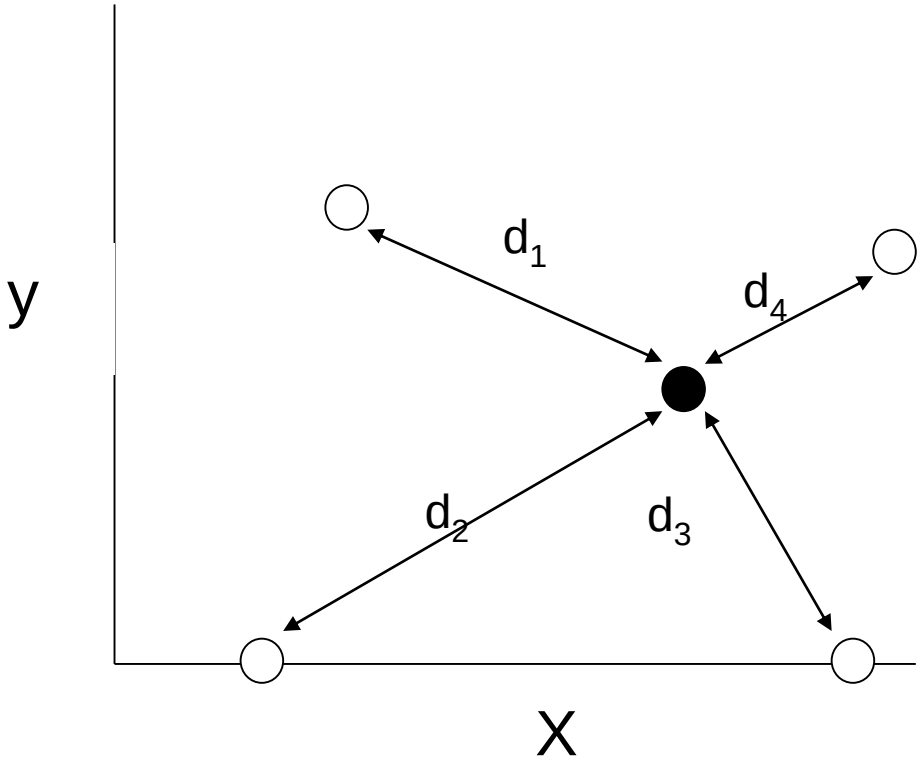
Principle: Surrounding gauges are weighted based on squares of reciprocals of distance away from the point of interest

Used for:

i. missing data estimation
ii. Un-gauged location data estimation

Missing data points or rainfall at ungauged locations (5/6)

Reciprocal Distance Method (2/3)



1. Distance away from the point of interest

$$d_i = \sqrt{x^2 + y^2}$$

2. X and y are the horizontal and vertical distances from the un-gauged (missing data) location where precipitation is being estimated

Missing data points or rainfall at ungauged locations (6/6)

Reciprocal Distance Method (3/3)

$$W_i = \frac{1}{d_i^2}$$

$$P_x = \frac{\sum_{i=1}^{n} P_i W_i}{\sum_{i=1}^{n} W_i}$$

Sample Question

Rain gauge	Monthly rainfall for June 2012 (mm)	Long term annual rainfall (mm)	Distance from P6 (km)
P_1	115	1300	7
P_2	156	1550	15
P_3	140	1350	20
P_4	150	1448	11
P_5	130	1288	4
P_6	X	1290	0
P_7	152	1524	14

Solution

Rain gauge	Monthly rainfall for June 2012 (mm) (Pi)	Distance from P6 (km) (d)	d^2	$W_{i=1/d^2}$	PiW_i
P1	115	7	49	0.02041	2.346939
P2	156	15	225	0.00444	0.693333
P3	140	20	400	0.0025	0.35
P4	150	11	121	0.00826	1.239669
P5	130	4	16	0.0625	8.125
P6	X	0	0		
P7	152	14	196	0.0051	0.77551
Total				0.103219	13.52899
Estimate	**140.5**			Px	**131.07**